# Three-Dimensional Lattice Boltzmann Formulation of the Magnetic Vector Potential for a Non-Collinear (MRSh) Ferrofluid

$\nabla^2 \boldsymbol{A} = -\mu_0 \nabla \times \boldsymbol{M}$, **MRT collision, single sub-step**


**Abstract**

The susceptibility closure $\nabla\cdot(\mu\nabla\psi) = 0$ presupposes $\boldsymbol{M} \parallel \boldsymbol{H}$ and is inadmissible once Martsenyuk–Raikher–Shliomis magnetisation dynamics is retained, because $\boldsymbol{M} \times \boldsymbol{H} \neq 0$ is the entire content of the rotational-viscosity effect. Replacing it by the scalar-potential Poisson equation $\nabla^2\psi = \nabla\cdot\boldsymbol{M}$ removes the contradiction but introduces a second one: a single-valued periodic $\psi$ carries zero mean field along any periodic direction, so an applied field parallel to a periodic axis cannot be represented on the lattice at all. This document develops the magnetic vector potential as the lattice variable instead. In three dimensions the Coulomb-gauge form $\nabla^2\boldsymbol{A} = -\mu_0\nabla\times\boldsymbol{M}$ decouples into three identical scalar Poisson equations, each solved by the same multiple-relaxation-time lattice Boltzmann scheme; $\nabla\cdot\boldsymbol{B} = 0$ then holds identically to round-off, and the applied field enters through boundary data rather than as an additive term. A representability theorem is proved for both potentials, showing them to be complementary. The three free constants of the scheme are determined rather than assumed, all boundary closures are derived from discrete balances, the gauge condition is shown to be preserved exactly by the component-wise solve, and the temporal response is characterised so that one sub-step per fluid step is provably sufficient. Error estimates are closed-form; numerical values appear only as sanity checks.


## I. GOVERNING EQUATIONS FOR THE MAGNETIC FIELD

### A. Magnetostatics and the failure of the susceptibility closure

For a non-conducting ferrofluid carrying no free currents,

$$\nabla \cdot \boldsymbol{B} = 0, \tag{1}$$

$$\nabla \times \boldsymbol{H} = 0, \tag{2}$$

where $\boldsymbol{B}$ and $\boldsymbol{H}$ are the magnetic induction and the magnetic field. In SI units

$$\boldsymbol{B} = \mu_0 \left(\boldsymbol{H} + \boldsymbol{M}\right). \tag{3}$$

The magnetisation is given by the MRSh effective-field law

$$\boldsymbol{M} = \phi\, \varphi_p M_s\, L(\xi_e)\, \frac{\boldsymbol{\xi}_e}{\xi_e}, \tag{4}$$

with $\phi$ the phase-field order parameter, $\varphi_p$ the particle volume fraction, and $\boldsymbol{\xi}_e$ the effective field obeying its own relaxation dynamics. Because (4) is non-collinear,

$$\boldsymbol{M} \times \boldsymbol{H} \neq 0 \qquad \Longrightarrow \qquad \boldsymbol{B} \neq \mu \boldsymbol{H} \ \text{ for any scalar } \mu, \tag{5}$$

so the closure $\nabla\cdot(\mu\nabla\psi) = 0$ is unavailable: it is equivalent to (1)–(3) only when $\boldsymbol{M} = \chi\boldsymbol{H}$.

### B. Vector-potential formulation

Equation (1) is satisfied identically by

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{6}$$

Substituting (6) and $\boldsymbol{H} = \boldsymbol{B}/\mu_0 - \boldsymbol{M}$ into (2),

$$\nabla \times (\nabla \times \boldsymbol{A}) = \mu_0 \nabla \times \boldsymbol{M} \qquad \Longleftrightarrow \qquad \nabla (\nabla \cdot \boldsymbol{A}) - \nabla^2 \boldsymbol{A} = \mu_0 \nabla \times \boldsymbol{M}. \tag{7}$$

Imposing the Coulomb gauge,

$$\nabla \cdot \boldsymbol{A} = 0, \tag{8}$$

reduces (7) to three uncoupled scalar Poisson equations,

$$\boxed{\nabla^2 A_\alpha = -\mu_0 (\nabla \times \boldsymbol{M})_\alpha \equiv -\mu_0 J_{b,\alpha}, \qquad \alpha = x, y, z} \tag{9}$$

$\boldsymbol{J}_b = \nabla \times \boldsymbol{M}$ being the bound-current density. The field follows as

$$\boldsymbol{H} = \frac{1}{\mu_0} \nabla \times \boldsymbol{A} - \boldsymbol{M}. \tag{10}$$

In the two-phase case, with $\boldsymbol{M} = \phi\, \boldsymbol{M}_f$,

$$\nabla \times \boldsymbol{M} = \phi \nabla \times \boldsymbol{M}_f + \nabla \phi \times \boldsymbol{M}_f, \tag{11}$$

the second term being the interfacial bound surface current.

### C. Preservation of the gauge

**Lemma 1** (Gauge harmonicity)**.** *If $A_\alpha$ satisfies (9) for each $\alpha$, then $\nabla \cdot \boldsymbol{A}$ is harmonic:*

$$\nabla^2 (\nabla \cdot \boldsymbol{A}) = -\mu_0 \nabla \cdot (\nabla \times \boldsymbol{M}) = 0. \tag{12}$$

Consequently $\nabla \cdot \boldsymbol{A}$ is determined entirely by its boundary data: the component-wise solve neither creates nor destroys gauge error. If the boundary data are consistent with (8), the gauge holds throughout; otherwise the residual satisfies, from (7),

$$\nabla \times \boldsymbol{H} = \frac{1}{\mu_0} \nabla (\nabla \cdot \boldsymbol{A}), \tag{13}$$

so a non-constant $\nabla \cdot \boldsymbol{A}$ appears directly as a spurious curl in $\boldsymbol{H}$. Note that $\nabla \cdot \boldsymbol{B} = 0$ holds identically and independently of the gauge, being a consequence of (6) alone.

## II. REPRESENTABILITY OF THE APPLIED FIELD

Whether a potential can carry a uniform applied field depends on the topology of the domain. The answer differs for the two potentials and dictates which one may be used.

**Theorem 2** (Scalar-potential obstruction)**.** *Let $\Omega$ be periodic in $\hat{\boldsymbol{e}}$ with period $L$, and let $\boldsymbol{H} = -\nabla \psi$ with $\psi$ single-valued and $\Omega$-periodic. Then*

$$\left\langle \boldsymbol{H} \cdot \hat{\boldsymbol{e}} \right\rangle_\Omega = 0. \tag{14}$$

*Proof.* For a line $C$ along $\hat{\boldsymbol{e}}$ closing through the periodic identification, $\oint_C \boldsymbol{H} \cdot d\boldsymbol{\ell} = -[\psi]_C = 0$. Dividing by $L$ and averaging over the transverse coordinates gives (14). □

**Theorem 3** (Vector-potential obstruction)**.** *Let $\Omega$ be periodic in two orthogonal directions $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$, and let $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ with $\boldsymbol{A}$ single-valued and $\Omega$-periodic. Then, with $\hat{\boldsymbol{n}} = \hat{\boldsymbol{e}}_1 \times \hat{\boldsymbol{e}}_2$,*

$$\left\langle \boldsymbol{B} \cdot \hat{\boldsymbol{n}} \right\rangle_\Omega = 0. \tag{15}$$

*Proof.* The cross-section $S$ spanned by $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ is a closed surface (a 2-torus), so $\partial S = \emptyset$ and Stokes' theorem gives $\int_S (\nabla \times \boldsymbol{A}) \cdot \hat{\boldsymbol{n}}\, dS = \oint_{\partial S} \boldsymbol{A} \cdot d\boldsymbol{\ell} = 0$. □

**Corollary 4** (Complementarity)**.** *A mean field along $\hat{\boldsymbol{e}}$ is representable by*

$$\psi \text{ iff } \hat{\boldsymbol{e}} \text{ is bounded}, \tag{16}$$

$$\boldsymbol{A} \text{ iff at least one direction perpendicular to } \hat{\boldsymbol{e}} \text{ is bounded}. \tag{17}$$

The two conditions are complementary, and $\boldsymbol{A}$ is the weaker requirement. For a channel periodic in $x$ and $z$ with walls normal to $y$:

| mean field along | $\hat{\boldsymbol{x}}$ | $\hat{\boldsymbol{y}}$ | $\hat{\boldsymbol{z}}$ |
|---|---|---|---|
| representable by $\psi$ | no | yes | no |
| representable by $\boldsymbol{A}$ | yes | no | yes |

Since the applied field in the Poiseuille benchmark is parallel to the flow, i.e. to a periodic direction, $\boldsymbol{A}$ is the admissible variable and $\psi$ is not. For a uniform $\boldsymbol{B}_0$ the potential that realises it is the linear ramp

$$\boldsymbol{A}_0 = \tfrac{1}{2}\, \boldsymbol{B}_0 \times \boldsymbol{x} \qquad \text{or, in the ramp gauge,} \qquad \boldsymbol{A}_0 = B_{0x}\, y\, \hat{\boldsymbol{z}} - B_{0z}\, y\, \hat{\boldsymbol{x}}, \tag{18}$$

both harmonic and divergence-free. The second form ramps only in $y$, i.e. only in the bounded direction, which is the practical statement of (17). Because $\nabla^2 \boldsymbol{A}_0 = 0$, adding or removing $\boldsymbol{A}_0$ changes neither the source in (9) nor the gauge; the split is exact and is used only when conditioning demands it.

## III. LATTICE BOLTZMANN MODEL FOR THE VECTOR-POTENTIAL EQUATION

### A. Pseudo-time embedding

Each component of (9) is embedded as the steady state of a diffusion equation in a fictitious time $t$, with a free parameter $\varepsilon$:

$$\frac{\partial A_\alpha}{\partial t} = \varepsilon \nabla^2 A_\alpha + S_\alpha, \qquad S_\alpha = \varepsilon\, \mu_0\, (\nabla \times \boldsymbol{M})_\alpha\,. \tag{19}$$

Setting $\partial_t A_\alpha = 0$ returns (9) for *any* $\varepsilon > 0$, because the same $\varepsilon$ multiplies both terms. If the $\varepsilon$ used in $S_\alpha$ differs from the diffusivity the lattice realises, the steady state becomes $\nabla^2 A_\alpha = -\mu_0(\varepsilon_{\text{code}}/\varepsilon_{\text{lattice}}) J_{b,\alpha}$, a silent multiplicative error on every field.

### B. Velocity set

Any DdQq set satisfying

$$\sum_k w_k = 1, \quad \sum_k w_k e_{k\alpha} = 0, \quad \sum_k w_k e_{k\alpha} e_{k\beta} = c_s^2 \delta_{\alpha\beta}, \quad \sum_k w_k e_{k\alpha} e_{k\beta} e_{k\gamma} = 0 \tag{20}$$

may be used. Since $A_\alpha$ obeys a pure diffusion equation, the minimal D3Q7 set is sufficient:

$$\boldsymbol{e}_k = \{(0,0,0), (\pm 1,0,0), (0,\pm 1,0), (0,0,\pm 1)\}, \qquad w_0 = \tfrac{1}{4},\ \ w_{1\ldots 6} = \tfrac{1}{8},\ \ c_s^2 = \tfrac{1}{4}. \tag{21}$$

D3Q15, D3Q19 and D3Q27 satisfy (20) equally and may be used if lattice uniformity with the momentum solver is preferred; all results below are stated in terms of $c_s^2$ and are therefore lattice-independent.

### C. MRT evolution

Three distribution functions $h_k^\alpha$, $\alpha = x, y, z$, evolve independently:

$$h_k^\alpha(\boldsymbol{x} + \boldsymbol{e}_k \Delta t, t + \Delta t) = h_k^\alpha(\boldsymbol{x}, t) - \left(\mathsf{M}^{-1} \mathsf{S}^h \mathsf{M}\right)_{k\ell} \left[h_\ell^\alpha(\boldsymbol{x}, t) - h_\ell^{\text{eq},\alpha}(\boldsymbol{x}, t)\right] + \Delta t\, \Phi_k^\alpha(\boldsymbol{x}, t), \tag{22}$$

with local equilibrium and discrete source

$$h_k^{\mathrm{eq},\alpha} = w_k A_\alpha, \qquad \Phi_k^\alpha = c_\Phi\, w_k\, S_\alpha. \tag{23}$$

The diagonal matrix is

$$\mathsf{S}^h = \mathrm{diag}\left(s_0^h, s_1^h, \ldots, s_{q-1}^h\right), \tag{24}$$

in which the rates acting on the three first-order moments are set equal,

$$s_{(x)}^h = s_{(y)}^h = s_{(z)}^h \equiv s_1^h, \qquad \frac{1}{s_1^h} = \frac{\varepsilon}{c_s^2 \Delta t} + 0.5, \tag{25}$$

and all remaining rates are free, subject to $0 < s_k^h < 2$. For D3Q7 the moment matrix may be taken as

$$\mathsf{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 \\ 6 & -1 & -1 & -1 & -1 & -1 & -1 \\ 0 & 2 & 2 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 1 & -1 & -1 \end{pmatrix}, \qquad \mathsf{S}^h = \mathrm{diag}\left(s_0^h, s_1^h, s_1^h, s_1^h, s_4^h, s_5^h, s_6^h\right). \tag{26}$$

Note the absence of $\mu$ in (25): equation (9) has constant coefficients, and $\varepsilon$ is a pseudo-time relaxation rate of dimension $\mathrm{L}^2\mathrm{T}^{-1}$. The macroscopic variables are

$$A_\alpha = \sum_k h_k^\alpha + a\, S_\alpha. \tag{27}$$

## IV. DETERMINATION OF THE SCHEME CONSTANTS

### A. MRT projection lemma

The analysis requires exactly two contractions of $\Lambda = \mathsf{M}^{-1}\mathsf{S}^h\mathsf{M}$. Both follow from the structure of $\mathsf{M}$, whose row 0 is $M_{0k} = 1$ and whose first-moment rows are $M_{(\alpha)k} = e_{k\alpha}/c$. Using $\sum_k M_{ak}(\mathsf{M}^{-1})_{kb} = \delta_{ab}$,

$$\sum_k \Lambda_{k\ell} = \sum_a s_a^h M_{a\ell} \sum_k (\mathsf{M}^{-1})_{ka} = \sum_a s_a^h M_{a\ell}\delta_{0a} = s_0^h, \tag{28}$$

$$\sum_k e_{k\alpha}\Lambda_{k\ell} = c\sum_a s_a^h M_{a\ell} \sum_k M_{(\alpha)k}(\mathsf{M}^{-1})_{ka} = s_1^h\, e_{\ell\alpha}. \tag{29}$$

That is: the zeroth moment relaxes at $s_0^h$ and the first moment at $s_1^h$; no other rate enters the recovered macroscopic equation. This holds for every lattice in (20) and reduces to BGK when $s_0^h = s_1^h$.

### B. Chapman–Enskog expansion

Dropping the component label, expanding the streaming operator and dividing by $\Delta t$ with $\mathscr{D}_k = \partial_t + \boldsymbol{e}_k\cdot\nabla$,

$$\mathscr{D}_k h_k + \frac{\Delta t}{2}\mathscr{D}_k^2 h_k + O(\Delta t^2) = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{\mathrm{neq}} + \Phi_k. \tag{30}$$

With $h_k = h_k^{(0)} + \delta h_k^{(1)} + \delta^2 h_k^{(2)}$, $\partial_t = \delta\partial_{t_1} + \delta^2\partial_{t_2}$, $\nabla = \delta\nabla_1$, $\Phi_k = \delta\Phi_k^{(1)}$, $S = \delta S_1$:

$$O(\delta^0): \quad h_k^{(0)} = w_k A, \tag{31}$$

$$O(\delta^1): \quad \Delta t\, \mathscr{D}_k^{(1)} h_k^{(0)} = -\Lambda_{k\ell} h_\ell^{(1)} + \Delta t\, \Phi_k^{(1)}, \tag{32}$$

$$O(\delta^2): \quad \partial_{t_2} h_k^{(0)} + \mathscr{D}_k^{(1)} h_k^{(1)} + \frac{\Delta t}{2}\left(\mathscr{D}_k^{(1)}\right)^2 h_k^{(0)} = -\frac{1}{\Delta t}\Lambda_{k\ell} h_\ell^{(2)}. \tag{33}$$

Applying $\mathscr{D}_k^{(1)}$ to (32) eliminates the second derivative from (33). Expanding (27) order by order gives the constraints $\sum_k h_k^{(1)} = -aS_1$, $\sum_k h_k^{(2)} = 0$. The first moment of (32) with (29) gives the diffusive flux

$$J_\alpha \equiv \sum_k e_{k\alpha} h_k^{(1)} = -\frac{\Delta t\, c_s^2}{s_1^h}\, \partial_{1\alpha} A, \tag{34}$$

and the zeroth moments of (32) and of the reduced (33), with (28), recombine to

$$\boxed{\frac{\partial A}{\partial t} = \underbrace{\Delta t\, c_s^2 \left(\frac{1}{s_1^h} - \frac{1}{2}\right)}_{\varepsilon} \nabla^2 A + \underbrace{\left(\frac{s_0^h a}{\Delta t} + c_\Phi\right)}_{\mathscr{C}} S + \underbrace{\left[a\left(1 - \frac{s_0^h}{2}\right) - \frac{\Delta t}{2} c_\Phi\right]}_{\mathscr{T}} \frac{\partial S}{\partial t}} \tag{35}$$

Matching (35) to (19) gives

$$\text{(C1)} \quad \varepsilon = \Delta t\, c_s^2 \left(\tau_h - \tfrac{1}{2}\right), \qquad \tau_h \equiv 1/s_1^h, \tag{36}$$

$$\text{(C2)} \quad \mathscr{C} = \frac{s_0^h a}{\Delta t} + c_\Phi = 1, \tag{37}$$

$$\text{(C3)} \quad \mathscr{T} = a\left(1 - \frac{s_0^h}{2}\right) - \frac{\Delta t}{2} c_\Phi = 0. \tag{38}$$

Equation (36) shows $\varepsilon$ depends on $s_1^h$ only. Conditions (C2) and (C3) together have the unique solution $a = \Delta t/2$, $c_\Phi = 1 - s_0^h/2$; but only (C2) fixes the steady state, so a second closure exists,

$$c_\Phi = 1, \qquad a = 0 \qquad \Longrightarrow \qquad \mathscr{C} = 1, \quad \mathscr{T} = -\frac{\Delta t}{2}, \tag{39}$$

exact at steady state with a residual $-\frac{\Delta t}{2}\partial_t S$. Section 9 shows (39) is preferable in the quasi-static regime, whereupon

$$A_\alpha = \sum_k h_k^\alpha. \tag{40}$$

Any pair violating (C2) rescales every field by $\mathscr{C} \neq 1$.

## V. RECOVERY OF THE INDUCTION AND THE FIELD

Since $\sum_k \boldsymbol{e}_k h_k^{\mathrm{eq}} = 0$, the first moment of $h^\alpha$ is purely non-equilibrium, and (34) inverts to give the gradient of each component directly:

$$\frac{\partial A_\alpha}{\partial x_\beta} = -\frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{k\beta}\, h_k^\alpha. \tag{41}$$

The induction is the curl assembled from these nine moments,

$$B_\alpha = \varepsilon_{\alpha\beta\gamma}\partial_\beta A_\gamma = -\frac{1}{\tau_h c_s^2 \Delta t}\, \varepsilon_{\alpha\beta\gamma} \sum_k e_{k\beta} h_k^\gamma, \tag{42}$$

written out as

$$B_x = -\frac{1}{\tau_h c_s^2 \Delta t}\left[\sum_k e_{ky} h_k^z - \sum_k e_{kz} h_k^y\right], \tag{43}$$

$$B_y = -\frac{1}{\tau_h c_s^2 \Delta t}\left[\sum_k e_{kz} h_k^x - \sum_k e_{kx} h_k^z\right], \tag{44}$$

$$B_z = -\frac{1}{\tau_h c_s^2 \Delta t}\left[\sum_k e_{kx} h_k^y - \sum_k e_{ky} h_k^x\right], \tag{45}$$

and finally

$$\boldsymbol{H} = \frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M}. \tag{46}$$

No auxiliary gradient stencil and no ghost nodes are required: the entire field, including its mean, comes from the lattice. The leading correction to (41), obtained from the first moment of the reduced (33), is proportional to $\partial_\beta S_\alpha$ and vanishes identically at convergence.

## VI. BOUNDARY CLOSURES FOR THE $h$ LATTICES

Let $\boldsymbol{x}_b$ be a boundary node, the wall lying halfway to its image, $\hat{\boldsymbol{n}}$ the outward unit normal,

$$K_{\text{in}} = \{k : \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} < 0\}, \qquad W_{\text{in}} = \sum_{k \in K_{\text{in}}} w_k, \qquad \boldsymbol{e}_{\sigma(k)} = \boldsymbol{e}_k - 2(\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}}, \qquad \boldsymbol{e}_{\bar{k}} = -\boldsymbol{e}_k. \tag{47}$$

**Lemma 5** (Wall weight)**.** *For any lattice satisfying* (20) *with* $\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} \in \{-1, 0, +1\}$,

$$W_{\text{in}} = \frac{c_s^2}{2}. \tag{48}$$

*Proof.* $\sum_k w_k (\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})^2 = c_s^2$, and the summand equals unity on $K_{\text{in}}$ and on its mirror image $K_{\text{out}}$ and zero elsewhere; the two sets carry equal weight. □ Hence $1/W_{\text{in}} = 6$ for D2Q9, D3Q15, D3Q19, D3Q27 and 8 for D3Q7.

### A. Neumann closure

Specular reflection alone, $h_k^\alpha = h_{\sigma(k)}^{*\alpha}$, transfers

$$\sum_{k \in K_{\text{in}}} h_k^\alpha - \sum_{k \in K_{\text{out}}} h_k^{*\alpha} = 0, \tag{49}$$

because $\sigma$ is a weight-preserving bijection $K_{\text{in}} \leftrightarrow K_{\text{out}}$; it therefore imposes $\partial A_\alpha / \partial n = 0$. A prescribed normal derivative $\gamma_\alpha$ is realised by

$$\boxed{h_k^\alpha(\boldsymbol{x}_b, t + \Delta t) = h_{\sigma(k)}^{*\alpha}(\boldsymbol{x}_b, t) + B_w^\alpha w_k, \qquad B_w^\alpha = \frac{\Delta t\, \varepsilon\, \gamma_\alpha}{W_{\text{in}}} = \frac{2\,\Delta t\, \varepsilon\, \gamma_\alpha}{c_s^2}, \qquad k \in K_{\text{in}}} \tag{50}$$

obtained by equating the injected mass $B_w^\alpha W_{\text{in}}$ to the Fickian flux $-\hat{\boldsymbol{n}} \cdot \boldsymbol{J} = \varepsilon\, \partial A_\alpha / \partial n$.

### B. Dirichlet closure

Where $A_\alpha$ is prescribed on the wall, the anti-bounce-back closure applies:

$$h_k^\alpha(\boldsymbol{x}_b, t + \Delta t) = -h_{\bar{k}}^{*\alpha}(\boldsymbol{x}_b, t) + 2 w_k A_{w\alpha}, \qquad k \in K_{\text{in}}. \tag{51}$$

For $h_k^{\text{eq}} = w_k A$, the combination $h_k + h_{\bar{k}}$ carries the even, potential-bearing part and $h_k - h_{\bar{k}}$ the odd, flux-bearing part; reversing the sign of the reflected population fixes the former at $2 w_k A_w$ while leaving the flux free.

### C. Physical boundary data

At a wall with vacuum outside carrying the applied field, continuity of the tangential $\boldsymbol{H}$ and of the normal $\boldsymbol{B}$ gives, with $\hat{\boldsymbol{t}}$ any unit tangent,

$$\hat{\boldsymbol{t}} \cdot \boldsymbol{B}\big|_{\text{wall}} = \mu_0 \left(\hat{\boldsymbol{t}} \cdot \boldsymbol{H}_0 + \hat{\boldsymbol{t}} \cdot \boldsymbol{M}\right), \qquad \hat{\boldsymbol{n}} \cdot \boldsymbol{B}\big|_{\text{wall}} = \mu_0\, \hat{\boldsymbol{n}} \cdot \boldsymbol{H}_0. \tag{52}$$

Since $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, the tangential induction involves the normal derivatives of the tangential components of $\boldsymbol{A}$ together with tangential derivatives of $A_n$. For a wall normal to $\hat{\boldsymbol{y}}$,

$$\frac{\partial A_z}{\partial y} = B_x + \frac{\partial A_y}{\partial z} = \mu_0 \left(H_{0x} + M_x\right) + \frac{\partial A_y}{\partial z}, \tag{53}$$

$$\frac{\partial A_x}{\partial y} = \frac{\partial A_y}{\partial x} - B_z = \frac{\partial A_y}{\partial x} - \mu_0 \left(H_{0z} + M_z\right), \tag{54}$$

$$\frac{\partial A_y}{\partial y} = -\frac{\partial A_x}{\partial x} - \frac{\partial A_z}{\partial z}, \tag{55}$$

the last being the Coulomb gauge (8) evaluated on the wall, which supplies the missing datum for $A_y$ and simultaneously enforces the gauge boundary condition required by the Lemma of Sec. I C. The remaining relation in (52),

$$\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} = \mu_0 H_{0y}, \tag{56}$$

is a condition on the wall-surface values of $A_x, A_z$ and fixes the residual gauge freedom tangential to the wall. All tangential derivatives in (53)–(56) are evaluated from the wall-layer values of $\boldsymbol{A}$ at the previous time level, so the three lattices are coupled only through their boundary data, never through their interiors. Wall values of $\boldsymbol{M}$ are extrapolated to second order,

$$M_\alpha^w = \tfrac{3}{2} M_\alpha\big|_{\text{1st node}} - \tfrac{1}{2} M_\alpha\big|_{\text{2nd node}}. \tag{57}$$

### D. Periodic directions

Populations stream unchanged. By Theorem 2 this is admissible precisely when the mean induction normal to the doubly-periodic plane vanishes; a non-zero $\langle B_n \rangle$ there would require a quasi-periodic closure $h_k^\alpha \leftarrow h_k^{*\alpha} + w_k \Delta A_{\text{per}}$, with the attendant conditioning penalty.

## VII. SOLVABILITY AND GAUGE CONSISTENCY

Summing (22) over $k$ and over the domain, and using (28) with (C2), the lattice mass of component $\alpha$ changes by $\Delta t\, S_\alpha$ per node and by $\Delta t\, \varepsilon \gamma_\alpha$ per Neumann boundary node. A steady state therefore exists, component by component, if and only if

$$-\mu_0 \sum_{\boldsymbol{x}} (\nabla \times \boldsymbol{M})_\alpha = \sum_{\text{faces}} \gamma_\alpha, \tag{58}$$

the discrete divergence theorem for $A_\alpha$. Residual stencil mismatch is removed by projecting the source onto the range of the discrete Laplacian,

$$\Delta_\alpha = -\mu_0 \sum_{\boldsymbol{x}} (\nabla \times \boldsymbol{M})_\alpha - \sum_{\text{faces}} \gamma_\alpha, \qquad S_\alpha \;\leftarrow\; S_\alpha - \frac{\varepsilon\, \Delta_\alpha}{N}, \tag{59}$$

$N$ the number of nodes. The projection is applied only where all faces of that component are Neumann or periodic; a single Dirichlet face renders the operator non-singular and (58) is then not required.

Gauge consistency is a separate condition. By the Lemma of Sec. I C the quantity $\nabla \cdot \boldsymbol{A}$ is harmonic and is therefore fixed entirely by its boundary data; the component-wise solve neither creates nor destroys it. Imposing (55) on every wall sets that data to zero, whence $\nabla \cdot \boldsymbol{A} \equiv 0$ and, by (13), $\nabla \times \boldsymbol{H} \equiv 0$. If a gauge residual is nonetheless detected, it is removed by one Poisson solve for a scalar $\lambda$,

$$\nabla^2 \lambda = \nabla \cdot \boldsymbol{A}, \qquad \boldsymbol{A} \;\leftarrow\; \boldsymbol{A} - \nabla \lambda, \tag{60}$$

which leaves $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ unchanged. This is the magnetostatic analogue of divergence cleaning in magnetohydrodynamics and reuses the same lattice.

## VIII. TEMPORAL RESPONSE AND THE SINGLE-SUB-STEP CONDITION

Linearising (19) for a Fourier mode of wavenumber $k$ driven at frequency $\omega$,

$$\hat{A} = \frac{\hat{S}}{\mathrm{i}\omega + \varepsilon k^2} = \frac{\hat{A}_{\text{qs}}}{1 + \mathrm{i}\,\omega\, \tau_{\text{relax}}}, \qquad \hat{A}_{\text{qs}} = \frac{\hat{S}}{\varepsilon k^2}, \tag{61}$$

$\hat{A}_{\text{qs}}$ being the true magnetostatic response. The lattice relaxation time is defined spectrally, $\tau_{\text{relax}} = -\Delta t / \ln |\lambda_2|$ with $\lambda_2$ the second-largest eigenvalue modulus of the one-step operator, and has two asymptotic branches,

$$\tau_{\text{relax}} \simeq \max\left( \frac{1}{\varepsilon k_{\min}^2},\; \tau_h \right), \qquad \tau_h = \frac{\varepsilon}{c_s^2 \Delta t} + \frac{1}{2}, \tag{62}$$

whose crossing gives a *minimum* in $\varepsilon$,

$$\varepsilon_{\text{opt}} \simeq \frac{c_s}{k_{\min}}. \tag{63}$$

Raising $\varepsilon$ beyond $\varepsilon_{\text{opt}}$ makes the solver slower, not faster: there is a floor on $\tau_{\text{relax}}$ that no choice of $\varepsilon$ defeats. One sub-step per fluid step is sufficient if and only if

$$\boxed{\omega\,\tau_{\text{relax}} \ll 1 \qquad \Longleftrightarrow \qquad \tau_B \gg (\omega\tau_B)\,\tau_{\text{relax}}} \tag{64}$$

since $\omega = (\omega\tau_B)/\tau_B$ and $\tau_B$ measured in lattice steps is a free nondimensionalisation choice. Condition (64) is the faithful lattice statement that magnetostatic relaxation is instantaneous relative to every physical time scale.

## IX. ERROR ANALYSIS

Four sources, each in closed form. Numerical values are sanity checks obtained from exact steady-state linear solves and eigenvalue computations of the scheme, not from production runs.

$$E_1 = \frac{\|A_h - A\|_\infty}{\|A\|_\infty} = C_1(k\Delta x)^2 + O\big((k\Delta x)^4\big), \qquad \text{spatial truncation (second order)} \tag{65}$$

$$C_1\big(a = \tfrac{\Delta t}{2}\big) - C_1(a=0) = \frac{\varepsilon\,\Delta t}{2}, \qquad \text{penalty of the half-source shift} \tag{66}$$

$$E_2^{\text{amp}} = 1 - \big[1 + (\omega\tau_{\text{relax}})^2\big]^{-1/2} \simeq \tfrac{1}{2}(\omega\tau_{\text{relax}})^2, \qquad \text{temporal lag, amplitude} \tag{67}$$

$$E_2^{\text{ph}} = \arctan(\omega\tau_{\text{relax}}), \qquad \text{temporal lag, phase} \tag{68}$$

$$E_3 = |\mathscr{C} - 1| = \left|\frac{s_0^h a}{\Delta t} + c_\Phi - 1\right|, \qquad \text{source normalisation} \tag{69}$$

$$E_4 = \frac{|\Delta_\alpha|/N}{|(\nabla\times\boldsymbol{M})_\alpha|}, \qquad \text{compatibility defect} \tag{70}$$

$$E_5 = \frac{\|\nabla\cdot\boldsymbol{A}\|_\infty}{\|\nabla\boldsymbol{A}\|_\infty}, \qquad \text{gauge residual, from (13)} \tag{71}$$

Combining (66) with the transient term of (39), the two admissible closures trade $\frac{\Delta t}{2}(\varepsilon k^2 - \omega)$, so that

$$\boxed{c_\Phi = 1,\ a = 0 \ \text{ is preferable whenever } \ \omega < \varepsilon k^2} \tag{72}$$

which is precisely the quasi-static condition (64). The half-source shift is required only when the drive approaches the lattice relaxation rate.

Two entries carry the argument of Sec. 2. The mean-field row confirms that the lattice reproduces an imposed uniform induction with no additive term, which the scalar potential cannot do along a periodic axis. The gauge row confirms Lemma 1: the divergence of the recovered $\boldsymbol{A}$ equals that of the input to twelve digits, so the solve is gauge-neutral and the gauge is decided entirely by (55). The spectral radius was exactly unity in every case tested — the conserved mode — so the floor in $\tau_{\text{relax}}$ is an accuracy limit, never a stability limit.

## X. EXACT TWO-DIMENSIONAL REDUCTION

If $\boldsymbol{M}$ is planar and independent of $z$, then $(\nabla\times\boldsymbol{M})_x = (\nabla\times\boldsymbol{M})_y = 0$ and (9) gives $A_x = A_y = 0$ up to harmonic terms. The system collapses to a single lattice,

$$\nabla^2 A_z = -\mu_0\left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}\right), \qquad \text{single scalar Poisson equation} \tag{73}$$

$$B_x = \frac{\partial A_z}{\partial y}, \qquad B_y = -\frac{\partial A_z}{\partial x}, \qquad \text{induction from one gradient} \tag{74}$$

$$\nabla\cdot\boldsymbol{A} = \frac{\partial A_z}{\partial z} \equiv 0, \qquad \text{gauge automatic} \tag{75}$$

| Quantity | Prediction | Measured | Test |
|---|---|---|---|
| $\varepsilon$ vs. (36) | $\Delta t c_s^2(1/s_1^h - \frac{1}{2})$, indep. of $s_0^h$ | ratio 0.998–1.0001 | $\varepsilon$-free source |
| Order of $E_1$ | 2 | 2.00 throughout | manufactured $A$ |
| $C_1$ gap, (66) | $\varepsilon\Delta t/2 = 2.5$ at $\varepsilon = 5$ | $3.771 - 1.045 = 2.73$ | manufactured $A$ |
| $E_3$, wrong $(c_\Phi, a)$ pairs | 0.500 | 0.495, 0.541 | manufactured $A$ |
| Neumann closure (50) | exact | rel. err $2.6 \times 10^{-15}$ | exact quadratic |
| Dirichlet closure (51) | 2nd order | 2.00; coefficient 1 gives 0.31 | exact quadratic |
| $W_{\text{in}} = c_s^2/2$ | exact | holds for Q7, Q9, Q15, Q19, Q27 | direct sum |
| 3D component solve | exact | $1.0 \times 10^{-17}$, $1.3 \times 10^{-18}$, $8.9 \times 10^{-16}$ | manufactured $\boldsymbol{A}$ |
| Gauge preservation, Lemma 1 | $\nabla\cdot\boldsymbol{A}$ unchanged | in $6.905 \times 10^{-5}$, out $6.905 \times 10^{-5}$ | manufactured $\boldsymbol{A}$ |
| $\nabla\cdot\boldsymbol{B}$ from (6) | 0 identically | $1.2 \times 10^{-16}$ | manufactured $\boldsymbol{A}$ |
| Mean field on the lattice | carried | $\langle B_x \rangle = 0.130000$ (imposed 0.13) | manufactured $\boldsymbol{A}$ |
| $\tau_{\text{relax}}$ minimum, (63) | at $\varepsilon \simeq c_s/k_{\min}$ | min at $\varepsilon = 5$, $\tau_{\text{relax}} = 30$ | spectral radius |

Table 1: Closed-form predictions against numerical measurement.

$$\gamma = \frac{\partial A_z}{\partial n} = \mu_0 \left( \hat{\boldsymbol{t}} \cdot \boldsymbol{H}_0 + \hat{\boldsymbol{t}} \cdot \boldsymbol{M} \right), \qquad \text{decoupled Neumann data} \tag{76}$$

The reduction is exact, not an approximation: $A_x = A_y = 0$ is an invariant subspace of (9). In two dimensions the gauge condition and the boundary coupling (53)–(56) disappear entirely, which is why the planar case costs one lattice rather than three and needs no gauge monitoring.

## XI. IMPLEMENTABLE FORM

Lattice units $\Delta x = \Delta t = 1$, $\mu_0 = 1$. For each $\alpha \in \{x, y, z\}$:

$$\tau_h = \frac{\varepsilon}{c_s^2} + \frac{1}{2}, \qquad s_1^h = 1/\tau_h, \qquad s_0^h = 1, \qquad \text{relaxation set} \tag{77}$$

$$J_{b\alpha} = (\nabla \times \boldsymbol{M})_\alpha \text{ (isotropic stencil)}, \qquad \text{bound current} \tag{78}$$

$$\Delta_\alpha = -\sum_{\boldsymbol{x}} J_{b\alpha} - \sum_{\text{faces}} \gamma_\alpha, \qquad S_\alpha = \varepsilon \left( J_{b\alpha} - \Delta_\alpha / N \right), \qquad \text{source with projection} \tag{79}$$

$$h_k^{\text{eq},\alpha} = w_k A_\alpha, \qquad A_\alpha = \sum_k h_k^\alpha, \qquad \text{equilibrium, moment} \tag{80}$$

$$h_k^{*\alpha} = h_k^\alpha - \left( \mathsf{M}^{-1} \mathsf{S}^h \mathsf{M} \right)_{k\ell} \left( h_\ell^\alpha - h_\ell^{\text{eq},\alpha} \right) + w_k S_\alpha, \qquad \text{MRT collision} \tag{81}$$

$$h_k^\alpha = h_{\sigma(k)}^{*\alpha} + \frac{2\varepsilon\gamma_\alpha}{c_s^2} w_k \quad (k \in K_{\text{in}}), \qquad \text{wall streaming} \tag{82}$$

$$\partial_\beta A_\alpha = -\frac{1}{\tau_h c_s^2} \sum_k e_{k\beta} h_k^\alpha, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}, \qquad \boldsymbol{H} = \boldsymbol{B}/\mu_0 - \boldsymbol{M}, \qquad \text{field recovery} \tag{83}$$

with $\gamma_\alpha$ from (53)–(55) and (57). Ordering within one time step, the three passes being separated because $\nabla \times \boldsymbol{M}$ and $\Delta_\alpha$ are non-local:

$$\underbrace{(83)}_{\boldsymbol{H} \text{ from } h} \rightarrow \underbrace{\text{MRSh}}_{\boldsymbol{M} \text{ from } \boldsymbol{H}, \boldsymbol{\Omega}} \rightarrow \underbrace{(78), (79)}_{J_b,\ \gamma,\ S} \rightarrow \underbrace{(80)}_{A,\ h^{\text{eq}}} \rightarrow \underbrace{(81), (82)}_{\text{collide, stream}} \tag{84}$$

The momentum and phase-field lattices keep their present position and are unaffected. Parameter selection follows (63) and (64); note also the identity

$$\frac{\varphi_p M_s}{H_0} = \frac{3\eta\varphi_p\xi_0}{\mu_0 H_0^2 \tau_B}, \tag{85}$$

which is an output of the parameter choice and must match the fluid being reproduced. Diagnostics to output each step: $\sum_{x,k} h_k^\alpha$ (bounded iff (58) holds), $E_4$, $E_5$, and $\omega\tau_{\text{relax}}$.

## XII. UNDERSTANDING THE EQUATIONS

### A. What the vector potential is, and why it replaces the scalar one

There are two ways to describe magnetised matter. The scalar potential $\psi$ treats it as *bound magnetic charge*: where the magnetisation converges or diverges, effective charge accumulates and generates a field, giving $\nabla^2\psi = \nabla\cdot\boldsymbol{M}$. The vector potential treats it as *bound current*: the ferrofluid is a sea of microscopic current loops, and where all the loops are aligned the neighbouring currents cancel exactly, so nothing happens. Only where the alignment *varies* — which is exactly $\nabla\times\boldsymbol{M}\neq 0$ — does the cancellation fail and a net current $\boldsymbol{J}_b$ survive. That leftover current is the source in (9), which is the textbook vector-potential equation $\nabla^2\boldsymbol{A} = -\mu_0\boldsymbol{J}_b$ with the bound current in place of a free one. In a sheared ferrofluid the magnetisation tilts by different amounts at different heights, so $\partial M_x/\partial y \neq 0$, and that gradient is the source.

Both descriptions are exact and equivalent in the continuum. They differ on a periodic lattice, and that difference is decisive.

### B. Why the mean field now lives on the lattice

Think of $\psi$ as height on a landscape and $\boldsymbol{H}$ as the downhill slope. A steady field along $x$ requires the ground to slope downhill all along $x$; but $x$ wraps, so after one lap you are back where you started, and you cannot have been descending the whole way. That is Theorem 1, and it is why a uniform applied field parallel to a periodic axis cannot be represented by $\psi$ at all and has to be added by hand afterwards.

The vector potential stores something different. For a field along $x$ it ramps along $y$: $A_z = \mu_0 H_{0x} y$. Since $y$ is bounded by walls, the ramp never has to close on itself, and the mean field is carried by the lattice like any other part of the solution. Theorem 2 gives the precise limit: $\boldsymbol{A}$ fails only when *both* directions perpendicular to the field are periodic, because then the cross-section is a closed surface and Stokes' theorem forces the flux through it to vanish. In a channel that excludes only the wall-normal mean field. The two potentials are complementary, and $\boldsymbol{A}$ is the less restrictive of the two.

### C. Why three lattices, and what the gauge is for

In three dimensions $\boldsymbol{A}$ has three components, so (9) is three scalar Poisson equations — but they are the *same* equation three times, so one solver is written once and called three times. The interiors never talk to each other; the coupling is confined to the boundary data (53)–(55).

The gauge is the price of the vector potential. $\boldsymbol{A}$ is not unique: adding the gradient of any scalar leaves $\boldsymbol{B} = \nabla\times\boldsymbol{A}$ unchanged. Something must remove that freedom, and (8) is the standard choice. It matters because the reduction from the true equation (7) to the three Poisson equations (9) used it: if $\nabla\cdot\boldsymbol{A}$ is not constant, (13) says a spurious curl appears in $\boldsymbol{H}$, which is unphysical.

Lemma 1 makes this manageable. Because $\nabla\cdot(\nabla\times\boldsymbol{M}) = 0$ identically, the quantity $\nabla\cdot\boldsymbol{A}$ obeys Laplace's equation, so it is fixed entirely by what happens on the boundary. The interior solve can neither create nor destroy it — confirmed numerically to twelve digits. Setting the gauge at the walls through (55) therefore sets it everywhere, and (60) is available as a fallback if monitoring shows drift.

One structural bonus comes free: $\nabla\cdot\boldsymbol{B} = 0$ holds to round-off no matter what, because $\boldsymbol{B}$ is a curl by construction. The scalar-potential route has to earn that; here it is an identity.

### D. What the lattice actually does

Equations (19)–(23) form a bucket-passing scheme. Each node holds $q$ numbers whose sum is $A_\alpha$. Collision nudges them toward the isotropic split $w_k A_\alpha$; streaming shovels them to neighbours. Left alone with no source this smears $A_\alpha$ until it satisfies $\nabla^2 A_\alpha = 0$. The source term injects bound current at each node, and the stationary state of that competition is exactly (9). The parameter $\varepsilon$ controls only how fast the balance is

reached, not where it settles — which is why (19) carries $\varepsilon$ in both terms, and why any mismatch between the two occurrences is a silent scaling error on the whole field.

### E. Why the three constants are not free

Equation (36) states what diffusivity the lattice actually realises for a given relaxation rate; the pairing must be exact. Equation (37) is the one that cannot be violated: break it and every field is multiplied by $\mathscr{C} \neq 1$, uniformly and invisibly. Equation (38) is optional — it removes a spurious $\partial_t S$ term — and (72) shows it is not worth taking in the slow-drive regime, because the shift it requires adds $\frac{\Delta t}{2}\varepsilon k^2$ of spatial error to remove only $\frac{\Delta t}{2}\omega$ of transient error. Note also that (25) carries no permeability: that factor belonged to $\nabla\cdot(\mu\nabla\psi) = 0$, where the diffusivity genuinely is $\varepsilon\mu$. Equation (9) has constant coefficients, so carrying $\mu_0$ there is dimensionally wrong.

### F. Why the wall term takes that form

Specular reflection maps incoming links to outgoing links one-to-one with equal weights, so (49) shows it moves exactly zero net $A_\alpha$ across the wall — it imposes $\partial A_\alpha/\partial n = 0$ whether or not that is wanted. Adding $B_w^\alpha w_k$ to the incoming links injects $B_w^\alpha W_{\text{in}}$ per step; setting that equal to the physical flux $\varepsilon\gamma_\alpha$ gives (50). The lattice-dependence collapses because $W_{\text{in}} = c_s^2/2$ exactly, so the closure is written once for every velocity set.

Since $\nabla A_\alpha$ is $\boldsymbol{B}$ rotated, prescribing $A_\alpha$ on a wall fixes the normal induction while prescribing $\partial A_\alpha/\partial n$ fixes the tangential induction. At a channel wall what is physically known is the tangential field — it equals the applied $\boldsymbol{H}_0$ by continuity — so Neumann data is the natural closure, and (53) is where the applied field enters the calculation.

### G. Why one sub-step suffices

The lattice cannot solve an elliptic equation instantaneously, so $\boldsymbol{A}$ lags $\boldsymbol{M}$ with time constant $\tau_{\text{relax}}$. Equation (61) is the exact first-order lag, and (64) says simply: make the drive slow compared with that lag. There is a floor on $\tau_{\text{relax}}$, because raising $\varepsilon$ accelerates the diffusive mode but decelerates the kinetic one, with the minimum at (63). That floor cannot be lowered — but it does not need to be, because $\omega = (\omega\tau_B)/\tau_B$ and $\tau_B$ in lattice steps is a free choice. The quasi-static condition is bought with $\tau_B$, not with sub-cycling. Usefully, (72) shows the same inequality also settles the choice of closure constants.

## XIII. VERIFICATION PROTOCOL

With $\boldsymbol{M}$ frozen and the momentum solver disabled:

1. **Manufactured solution, per component.** Choose $A_\alpha$ with zero wall slope, set $J_{b\alpha} = -\nabla^2 A_\alpha/\mu_0$, confirm $E_1 \propto \Delta x^2$ and, critically, that the converged $A_\alpha$ is *independent of* $\varepsilon$. This single test validates (36) and (37) simultaneously.
2. **Non-zero wall flux.** Repeat with $\gamma_\alpha \neq 0$ to exercise (82); $\nabla^2 A = \text{const}$ with prescribed end slopes is quadratic and must be reproduced to round-off.
3. **Divergence-free manufactured $\boldsymbol{A}$.** Build $\boldsymbol{A} = \nabla \times \boldsymbol{P}$ with the same discrete operators, solve all three components, and check that $\nabla\cdot\boldsymbol{A}$ is unchanged (Lemma 1) and that $\nabla\cdot\boldsymbol{B}$ is at round-off.
4. **Mean-field test.** Add a ramp $A_z = B_{0x}y$ and verify $\langle B_x\rangle = B_{0x}$ with no additive term anywhere in the code. This is the test that distinguishes this formulation from the scalar-potential one.
5. **Uniformly magnetised slab.** $\boldsymbol{M} = M_0\hat{\boldsymbol{y}}$ gives $H_y = -M_0$ exactly and $H_x$ undisturbed; a returned $\mathscr{C}M_0$ with $\mathscr{C} \neq 1$ localises the failure to (37).
6. **Solvability and gauge.** Verify $\sum_{\boldsymbol{x},k} h_k^\alpha$ bounded, $E_4$ and $E_5$ at round-off. Linear growth of the first indicates (58) is violated; growth of $E_5$ indicates (55) is not being applied.

7. **Planar consistency.** With planar $\boldsymbol{M}$ and no $z$-dependence, confirm $A_x = A_y = 0$ to round-off, i.e. that the reduction of Sec. X is realised discretely and not merely in the continuum.